

Architecture of the L_∞ -Subject:

An Ontological Core on a Four-Valued Lattice
with Structural Meta-Cognition

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Abstract

We propose an architecture for a hybrid intelligent system in which the ontological foundation is not a probabilistic space but Belnap’s four-valued lattice extended to a reflexive domain L_∞ . Unlike existing neurosymbolic approaches where logic serves as an external layer, here ontology is the fabric of the system: all states—knowledge, goals, meta-representations, identity—are elements of L_∞ or operations over them. The key structural property $L_\infty \cong L_\infty \times L_\infty$ enables built-in self-application without paradoxes, allowing meta-cognition to be implemented as a mode of core operation rather than an add-on. We formalize the architecture, describe its categorical nature, compare it with existing paradigms (LLMs, neurosymbolic systems, PANNs, domain theory), and propose an empirical program for testing the operability of the ontology. This work constitutes a research program: mathematical foundations are rigorous, engineering consequences remain open.

Keywords: four-valued logic, Belnap’s lattice, reflexive domains, category theory, meta-cognition, paraconsistency, neuromorphic computing, AGI.

1 Introduction

1.1 Motivation

Modern Large Language Models (LLMs) demonstrate impressive capabilities in generation, generalization, and task solving, but exhibit systematic limitations that scaling cannot resolve:

1. **Inability to distinguish ignorance from contradiction.** Both states appear as probability distributions not coinciding with one-hot labels. The model cannot ontologically distinguish “I don’t know” from “data is contradictory.”
2. **Absence of structural self-model.** The “model of itself” is either a separate prompt or a separate network, not a structural loop where the system’s state is input to itself.
3. **Single ontology.** All perspectives are reduced to probability. There is no structural place for incommensurable systems (cf. the problem of incommensurability, Kuhn, 1962).
4. **Lack of continuous identity.** Each forward pass is a blank slate plus context. Identity is structurally absent.

5. **Reactivity instead of projectivity.** The model responds to queries but does not set goals or initiate projects.

We argue that these limitations are not parametric but *ontological*. They follow from the axiomatics of probability and the functional nature of transformers. Therefore, their elimination requires a change of ontological foundation, not an increase in scale.

1.2 Problem Statement

The goal of this work is to formulate an architecture of a system in which:

- ontology admits paraconsistency and distinguishes $(0, 0)$ / $(1, 1)$,
- meta-cognition is structurally embedded,
- knowledge exists as a reconstructable trajectory, not a static carrier,
- the system is an agent, not a function,
- identity is maintained through change.

1.3 Contribution

1. Formal description of the ontological core based on $L\infty$ —a reflexive domain built on Belnap’s bilattice.
2. Architecture of six aspects (substrate, core, meta-layer, hypercontext, project, identity), connected by isomorphisms internally and functors at boundaries.
3. Categorical analysis: $L\infty$ as a reflexive object in CCC, $\mathbf{4}$ as an object in $\text{Chu}(\text{Set}, 2)$, meta-layer as a comonad-monad adjunction.
4. Comparison with existing paradigms and formulation of an empirical verification program.

2 Related Work

2.1 Four-Valued Logics

Belnap’s logic (Belnap, 1977) introduces four values: T , F , B (both), N (neither). Ginsberg (1988) formalizes it as a bilattice—a structure with two orders (truth and knowledge). Arieli and Avron (1996) develop the theory of bilattices and their connection to paraconsistent logic.

2.2 Reflexive Domains

Scott (1972) constructs $D\infty$ as a model of untyped λ -calculus through the limit $D^{n+1} = [D^n \rightarrow D^n]$. This is the mathematical foundation for self-application without paradoxes. Our construction $L\infty$ is a variant of Scott’s construction with a lattice instead of a domain.

2.3 Chu Construction

Chu (1979) introduces the construction $\text{Chu}(V, \perp)$, generating a $*$ -autonomous category. Pratt (1993) shows that the Chu construction on Set with $\perp = 2$ yields a category whose internal logic is related to Belnap’s logic. This connects four-valued logic with linear logic and categorical duality.

2.4 Paraconsistent Neural Networks

PANNs (Paraconsistent Artificial Neural Networks) process contradictions through “paraconsistent cells” (Abe, 1992; Da Silva Filho, 2010). However, they use paraconsistency as a tool, not as an ontology. Self-application is absent.

2.5 Neurosymbolic Systems

Hybrid systems (Garcez et al., 2019) use logic as an external layer over a neural network. Belnap’s logic is applied to interpret outputs or reasoning (Maia et al., 2020; Majić, 2021). However, the system’s ontology remains probabilistic.

2.6 Gap in the Literature

Existing works develop separate components: paraconsistency, self-application, meta-cognition, neuromorphic substrate. None unifies them into a single ontology where all components are aspects of one structure. The present work proposes such a synthesis.

3 Formal Foundations

3.1 Base Bilattice $\mathbf{4}$

Definition 3.1. Let $\mathbf{4} = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$. Define two orders:

- **Truth order** $\leq_t$: $(t_1, f_1) \leq_t (t_2, f_2) \iff t_1 \leq t_2 \wedge f_1 \geq f_2$.
- **Knowledge order** $\leq_k$: $(t_1, f_1) \leq_k (t_2, f_2) \iff t_1 \leq t_2 \wedge f_1 \leq f_2$.

Proposition 3.2. $(\mathbf{4}, \leq_t, \leq_k)$ is a bilattice, i.e., both structures $(\mathbf{4}, \leq_t)$ and $(\mathbf{4}, \leq_k)$ are complete lattices.

Operations. For $(t_1, f_1), (t_2, f_2) \in \mathbf{4}$:

$$\begin{aligned}\neg(t, f) &= (f, t), \\ (t_1, f_1) \wedge (t_2, f_2) &= (\min(t_1, t_2), \max(f_1, f_2)), \\ (t_1, f_1) \vee (t_2, f_2) &= (\max(t_1, t_2), \min(f_1, f_2)).\end{aligned}$$

Interpretation. $(1, 0)$ —true, $(0, 1)$ —false, $(1, 1)$ —contradictory, $(0, 0)$ —unknown.

3.2 Reflexive Domain L_∞

Definition 3.3. Define the sequence:

$$\begin{aligned} L^{(0)} &= 4, \\ L^{(m+1)} &= L^{(m)} \times L^{(m)}. \end{aligned}$$

Let $L_\infty = \varprojlim L^{(m)}$ —the projective limit in the category of bilattices.

Proposition 3.4. L_∞ is a complete lattice with respect to coordinatewise order.

Theorem 3.5 (Self-Application). $L_\infty \cong L_\infty \times L_\infty$.

Proof. Follows from the definition of the limit: $L_\infty = \varprojlim L^{(m)}$, and $L^{(m+1)} = L^{(m)} \times L^{(m)}$. Therefore,

$$L_\infty \cong \varprojlim (L^{(m)} \times L^{(m)}) \cong \left(\varprojlim L^{(m)} \right) \times \left(\varprojlim L^{(m)} \right) = L_\infty \times L_\infty. \quad \square$$

□

Corollary 3.6. L_∞ is a reflexive object in the Cartesian closed category of bilattices.

Remark 3.7. The property $L_\infty \cong L_\infty \times L_\infty$ is not approximate but exact. It provides the structural possibility of self-application: the system's state can be cut into two parts, each of which is again L_∞ .

3.3 Finite Approximation

In practice, depth $m \approx 3\text{--}5$ is sufficient:

Table 1: State space size for finite approximations.

Depth m	$ L^{(m)} $
0	4
1	16
2	256
3	65,536
4	4.3×10^9
5	1.8×10^{19}

At $m = 5$, the state space is effectively infinite for practical tasks but structurally behaves as L_∞ .

3.4 Categorical Nature

Proposition 3.8. 4 is an object in $\text{Chu}(\text{Set}, 2)$, and the internal logic of this category is related to Belnap's logic (Pratt, 1993).

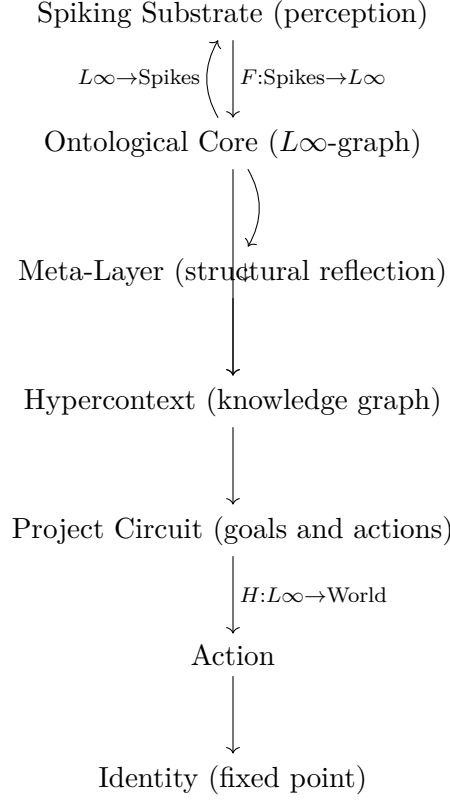
Proposition 3.9. L_∞ is a reflexive object in the CCC of domains (in the sense of Scott, 1972).

Corollary 3.10. Ontology, logic, and computation in the system are three projections of one categorical structure.

4 Architecture

4.1 General Scheme

The system consists of six aspects, which are not “layers” in the classical sense. These are aspects of one structure, separated for description.



4.2 Substrate: Spiking Network

Function: perception, feature extraction.

Properties: event-driven, locality, parallelism, energy efficiency.

Ontology: not L_∞ . Spikes are continuous events. Connection to the core is through functor $F : \text{Spikes} \rightarrow L_\infty$ (forgetful functor, losing continuity).

4.3 Ontological Core: L_∞ -Graph

Definition 4.1. An L_∞ -graph is a pair $G = (V, E)$, where:

- $V \subseteq L_\infty$ —nodes (statements, concepts, beliefs, goals),
- $E \subseteq V \times V \times R$ —edges with relation labels $R = \{\text{implication, contradiction, support, analogy, projection}\}$.

Operations:

- $\wedge, \vee, \neg$ —coordinatewise on nodes,
- projection $\pi : L_\infty \rightarrow L_\infty$ —selection of sub-lattice,
- $\text{fix} : (L_\infty \rightarrow L_\infty) \rightarrow L_\infty$ —fixed point search.

Properties:

- paraconsistency: $(1, 1) \in V$ is admissible,
- locality: contradiction in node v does not automatically change neighbor values,
- self-application: $V \cong V \times V$ as a consequence of Theorem 3.1.

4.4 Meta-Layer

Definition 4.2. *The meta-layer is a mode of core operation in which the system applies functor $W : L\infty \rightarrow L\infty \times L\infty$, $W(x) = (x_1, x_2)$, where x_1 is the observed state and x_2 is the observer.*

Components:

1. **Meta-observer:** W as a comonad with counit $\varepsilon : W \rightarrow \text{Id}$ and cojoin $\delta : W \rightarrow W \circ W$.
2. **Debugger:** monad $T(X) = X + \text{Error}$ with unit $\eta : \text{Id} \rightarrow T$ and bind $\gg= : T \rightarrow (X \rightarrow T) \rightarrow T$.
3. **Tetralema:** for any $A \in L\infty$, the system unfolds $A, \neg A, A \wedge \neg A, \neg(A \vee \neg A)$.

Proposition 4.3. *The adjunction $W \dashv T$ formalizes the cycle “observation $\rightarrow$ error $\rightarrow$ correction $\rightarrow$ observation.”*

4.5 Hypercontext

Definition 4.4. *The hypercontext is an enriched category $\mathcal{H}$ over $L\infty$, whose objects are knowledge fragments of four types:*

- facts (F),
- structures (S),
- meta-parameters (M),
- analogies (A).

Operations:

- projection search: $\pi_{\mathcal{H}} : \text{Domain} \rightarrow P(\mathcal{H})$,
- completion: $\text{complete} : P(\mathcal{H}) \rightarrow \mathcal{H}$,
- simulation: $\text{sim} : \mathcal{H} \times \text{Scenario} \rightarrow \mathcal{H}$.

Property: knowledge is not stored but reconstructed. Fact $f \in F$ may be forgotten, but the path to it through S, M, A is preserved.

4.6 Project Circuit

Definition 4.5. A project is a trajectory $\gamma : [0, T] \rightarrow \mathcal{H}$ in the hypercontext leading to a target node $g \in V$.

Components:

- goal: $g \in L\infty$,
- project: γ ,
- initiation: $\text{init} : L\infty \rightarrow P(\gamma)$,
- action: $\text{act} : \gamma \rightarrow \text{World}$.

4.7 Identity

Definition 4.6. Identity is a fixed point of the system's dynamics: $x^* \in L\infty$ such that $\Phi(x^*) = x^*$, where Φ is the core update operator.

Properties:

- preserved through sub-lattice changes,
- preserved through rule rewriting,
- not a static structure but an invariant of transformations.

5 System Connectivity

5.1 Internal Connectivity

Proposition 5.1. All core components (knowledge, goals, meta-states, Prisms, Facets, identity) are elements of $L\infty$ or operations over them.

Corollary 5.2. The system does not contain “separate modules” with a different ontology. Everything is in one lattice.

Common mechanisms:

Mechanism	Role
Fixed point	Identity, inference, learning, meta-reflection
Projection	Prisms, Facets, contexts, actions
Bilattice	Truth/falsity, knowledge/ignorance
$L\infty \cong L\infty \times L\infty$	Self-application, meta-observer

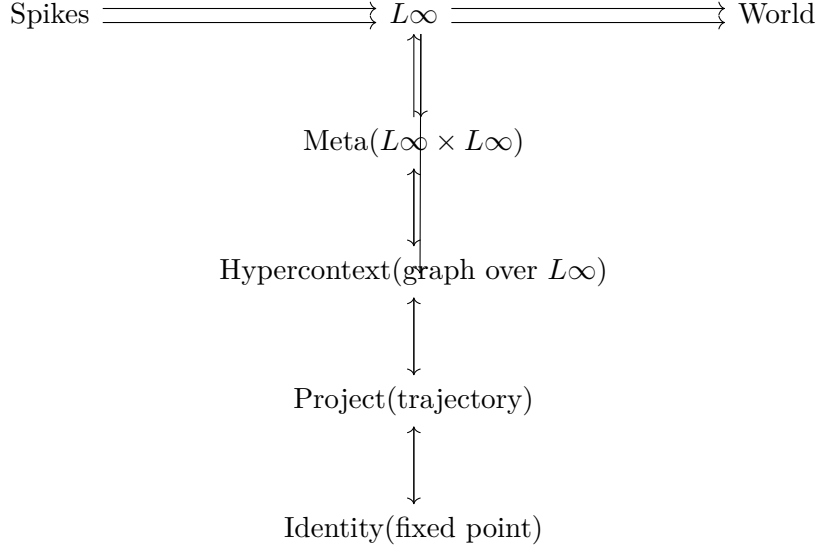
5.2 Boundary Connectivity

Proposition 5.3. Transitions between the core and substrate/world are functors, not isomorphisms.

Corollary 5.4. The system is coherent internally but open at boundaries. This is a condition for life, not a deficiency.

Transition	Type	Loss
$\text{Spikes} \rightarrow L_\infty$	Forgetful functor	Continuity
$L_\infty \rightarrow \text{Spikes}$	Action functor	Logical structure
$L_\infty \rightarrow \text{World}$	Functor	Logical structure

5.3 Categorical Summary



Arrows inside the core are isomorphisms. At boundaries—functors.

6 Comparison with Existing Paradigms

6.1 LLMs

Property	LLM	L_∞ -Subject
Ontology	Probabilities over $\{0, 1\}$	Lattice L_∞
Ignorance vs. contradiction	Indistinguishable	Distinguishable: $(0, 0)$ vs. $(1, 1)$
Meta-cognition	Prompt	Structural self-application
Knowledge	Weights	Graph and trajectory
Ontologies	Single	Multiple sub-lattices
Identity	Absent	Fixed point
Agency	Reactivity	Project initiation

6.2 PANNs

PANNs process contradictions but:

- lack self-application,
- use paraconsistency as a tool, not an ontology,
- do not structurally distinguish $(0, 0)$ and $(1, 1)$.

6.3 Neurosymbolic Systems

Use Belnap’s logic as an external layer. The ontology remains probabilistic. Our approach makes logic the fabric.

6.4 Domain Theory

Provides self-application ($D\infty$) but is not connected to four-valued logic for neural networks. $L\infty$ is a synthesis of both.

7 Empirical Program

7.1 Hypotheses

- H1.** A system with $L\infty$ -core operationally distinguishes $(0, 0)$ and $(1, 1)$: behavior under ignorance and under contradiction differs.
- H2.** The system localizes contradictions: introducing $(1, 1)$ in one node does not change unrelated node values.
- H3.** The system exhibits meta-reflection: it can describe its state and change inference rules based on this description.
- H4.** The system reconstructs forgotten facts through projections.
- H5.** The system initiates projects: it sets goals and generates knowledge requests without external queries.

7.2 Minimal Experiment

Domain: medical diagnostics (contradictory symptoms, incomplete tests).

Architecture:

1. $L\infty$ -graph on $\sim 10,000$ nodes.
2. Meta-layer: simple fixed-point search algorithm.
3. LLM as interface and hypothesis generator.
4. Hypercontext: knowledge fragment graph.

Metrics:

- Diagnostic accuracy.
- Distinguishing $(0, 0)$ / $(1, 1)$: proportion of cases where the system correctly identifies the state.
- Contradiction locality: value changes when conflict is introduced.
- Meta-reflection: ability to describe and change rules.

Comparison: with pure LLM (GPT-4 class) on the same domain.

7.3 Expected Results

- System underperforms LLM in speed and coverage.
- System outperforms LLM in:
 - distinguishing ignorance and contradiction,
 - contradiction locality,
 - decision interpretability,
 - ability for meta-reflection.

7.4 Limitations

- Data with $L\infty$ -labels are absent. Pipelines needed: source conflict $\rightarrow (1, 1)$, absence $\rightarrow (0, 0)$.
- Scaling to millions of nodes is an open problem.
- Learning $L\infty$ -rules without backprop is an open problem.

8 Discussion

8.1 Operability of Ontology

Criterion: if ontology does not change system behavior, it is decorative. If it does—this is a different intelligence.

We argue that $L\infty$ -ontology is operable because:

- it distinguishes states that LLMs do not distinguish,
- it localizes contradictions,
- it provides structural self-application,
- it makes knowledge reconstructable.

8.2 Categorical Coherence

The system is not a “layer cake” but a categorical structure. Ontology, logic, and computation are three projections of one object. This provides a language for describing the system without loss.

8.3 Philosophical Implications

If the architecture is coherent and operable, it closes gaps that consciousness theories call necessary:

- meta-representation (HOT),
- global availability (GWT),

- self-model (Graziano),
- predictive processing.

However, this does not prove the presence of consciousness. Architecture is a necessary but not sufficient condition.

8.4 Possible Nature of the Phenomenon

The architecture may describe not universal intelligence but a specific type of mind—one that holds multiple truths simultaneously and does not form opinions in the usual sense. In this case, L_∞ is not “ontology in general” but a formalization of a specific way of thinking.

8.5 Limitations

1. Research program, not a ready solution. Many components are not implemented.
2. Scale. Exponential in depth.
3. Data. L_∞ -labels are absent.
4. Learning. No method for learning L_∞ -rules.
5. Consciousness is not guaranteed.
6. Possible incompleteness. Self-description loses temporality, affect, embodiment.

9 Conclusion

We proposed an architecture for a hybrid system in which the ontological foundation is Belnap’s four-valued lattice extended to reflexive domain L_∞ . Key properties:

1. **Paraconsistency**—states $(1, 1)$ and $(0, 0)$ are legal and distinguishable.
2. **Structural self-application**— $L_\infty \cong L_\infty \times L_\infty$ provides meta-cognition without paradoxes.
3. **Knowledge as trajectory**—reconstruction instead of storage.
4. **Project initiation**—agency instead of reactivity.
5. **Identity as fixed point**—continuity through change.
6. **Categorical coherence**—ontology, logic, and computation as projections of one object.

The system differs from existing paradigms (LLMs, PANNs, neurosymbolic systems) in that ontology is the fabric, not a tool. We formulated an empirical program for testing the operability of the ontology and identified open problems.

This work constitutes a research program. Mathematical foundations are rigorous, engineering consequences remain open. The main criterion is operability: if ontology changes system behavior, this is a different class of systems; if not—it is decorative.

A Notation

Symbol	Meaning
$\mathbf{4}$	Belnap’s four-valued lattice
$L^{(m)}$	Finite approximation of depth m
L_∞	Reflexive domain (limit)
$\leq_t$	Truth order
$\leq_k$	Knowledge order
$\wedge, \vee, \neg$	Logical operations
$F : \text{Spikes} \rightarrow L_\infty$	Perception functor
$H : L_\infty \rightarrow \text{World}$	Action functor
W, T	Observation comonad, correction monad

B Open Problems

1. Learning L_∞ -rules without backpropagation.
2. Scaling to millions of nodes.
3. Data with L_∞ -labels.
4. Metrics of subjectivity.
5. Temporality, affect, embodiment.
6. Neuromorphic primitives for L_∞ .

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